CII SR -14th Edition Kaizen Compétition 2019

A Kaizen Presentation By

EVEREST INDUSTRIES LIMITED

(Podanur Works – Coimbatore)

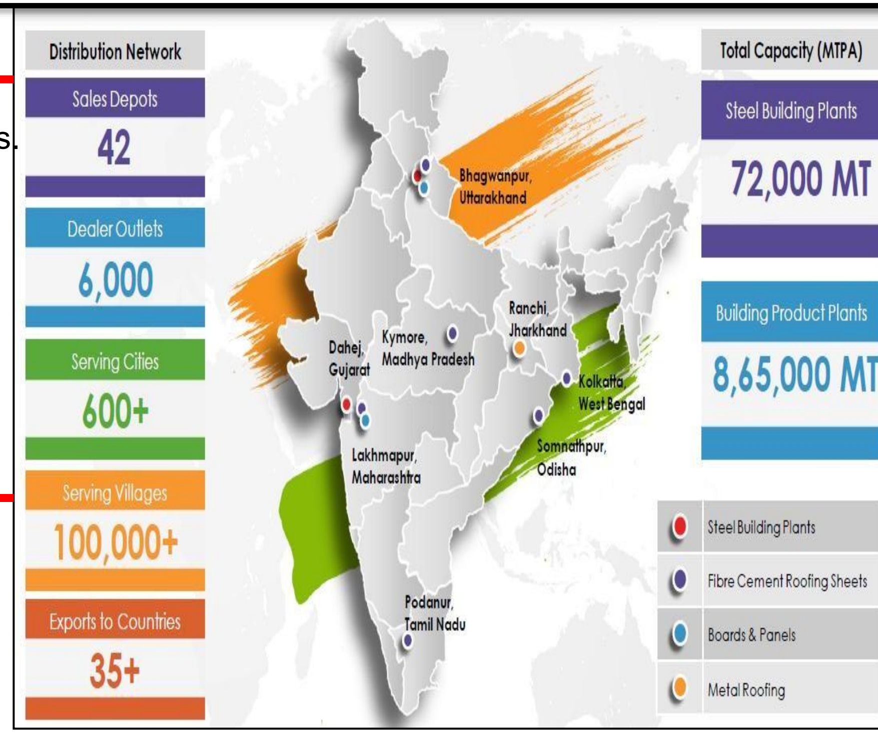
Company Profile

Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

Our Business:

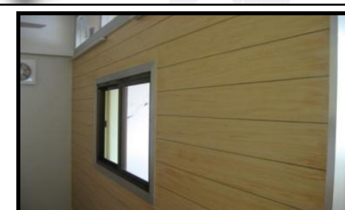
- **Building Products (63%) –** include fiber cement roofing sheets, fiber cement boards, solid wall panels.
- **Steel buildings (37%) –** offers customized building solutions like Pre-Engineered Steel buildings, metal roofing sheets and cladding.




Fibre Cement Roofing Sheets


Hi-Tech Roofing Sheets


Super Sheets


Cement Planks


Fibre Cement Boards


Rapicon Walls


Smart Steel Buildings

Our Esteemed Customers

EVEREST

Reliance
Industries Limited

Simplex Infrastructures Limited

Walmart
Save money. Live better.

IFB

ACC

Mahindra

HYATT
CAPITAL GATE
ABU DHABI

Pidilite

TATA

Whirlpool
CORPORATION

RELIANCE

INDIAN RAILWAYS

Marriott
HOTELS & RESORTS

ITC Limited

BRITANNIA
Eat Healthy, Think Better

DLF

DLF
BUILDING INDIA

एन बी सी सी
NBCC
A Navratna CPSE

Cipla

Coca-Cola

VIDEOCON

Kaizen Theme

Kaizen Theme: To improve productivity by reducing Machine cycle time.

Implemented Area: Sheeting Machine-1 Shop Floor, Podanur Works, Coimbatore

Implementation Date: 12th December 2018

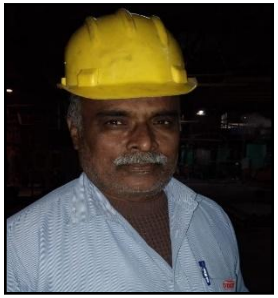
Cross Functional Team:

Mr. N. Karthikeyan
(Engineer - Production)

Team Leader
Mr. M. Vijay Kumar
(Sr. Engineer, Elect Maintenance.)

Mr. N. Anil Kumar
Engineer - QS & BE

Mr. Srinivasagam
(Electrician)

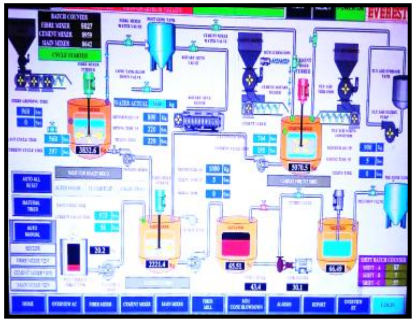


Mr. Narayana samy
(Board Operator)

Problem / Present Status

EVEREST

Process Flow:

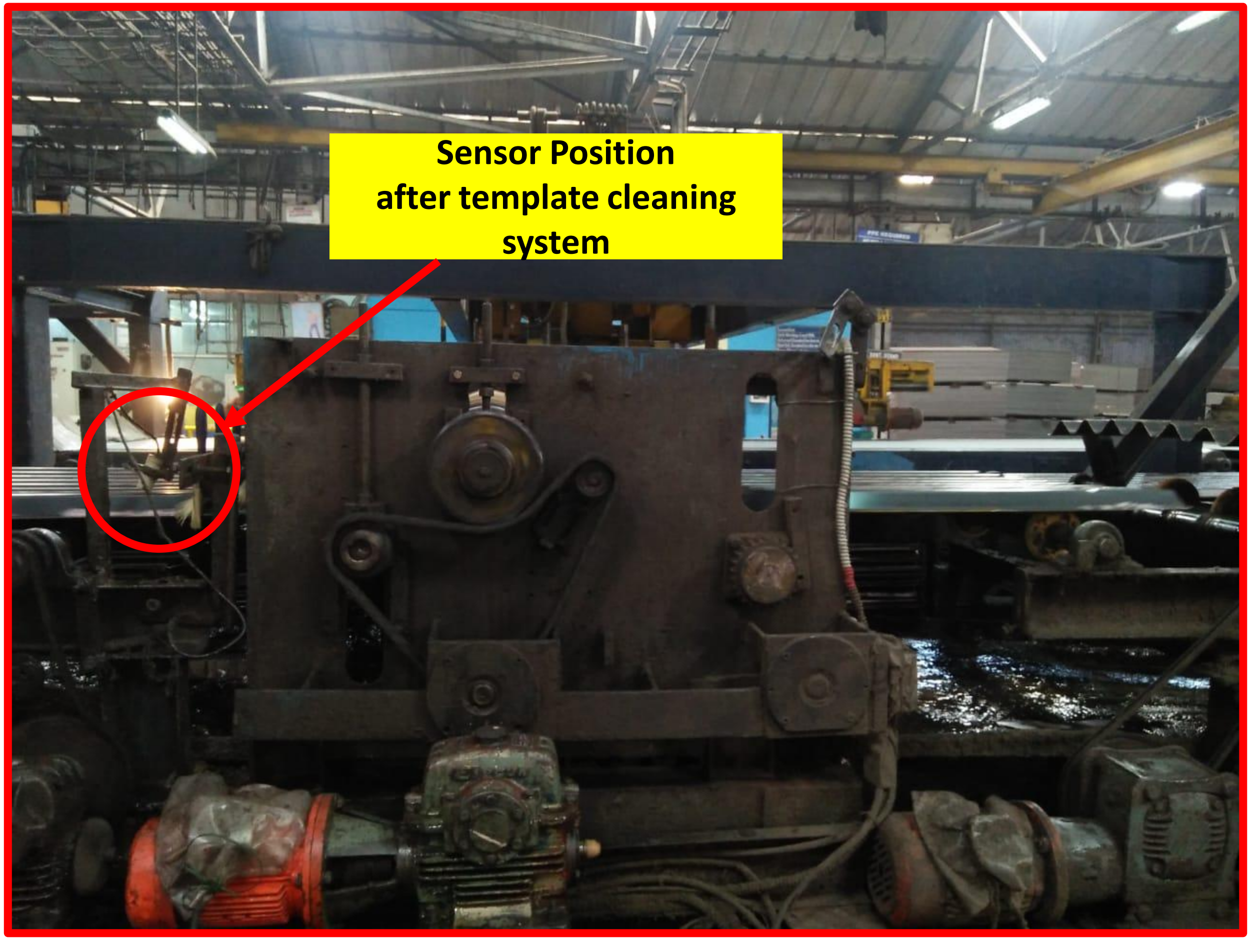


RAW MATERIAL PREPARATION → VATS → SHEET FORMING → SHEET HANDLING / PILING MACHINE

↓

Oiling System ← DE-PILING ← PRE CURING

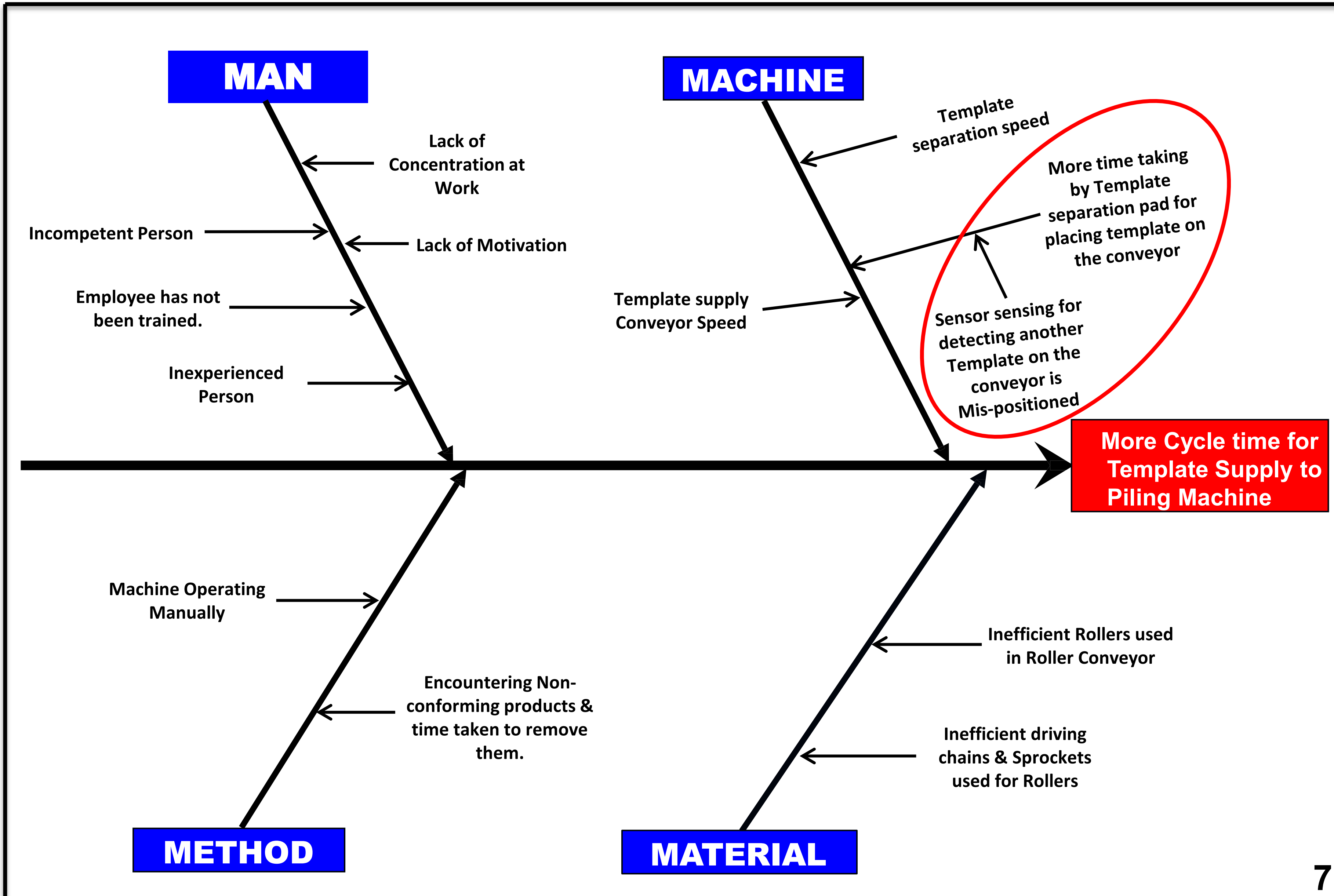
Problem / Present Status



Problem/ Present Status :

Cycle time of the Template supply system to Pilling machine is 16 seconds. Because of this we can't able to reduce the Machine speed below 16 seconds.

Cause & Effect Analysis



Root Cause Analysis & Idea Generation

Problem: More cycle time for Template supply to Piling machine.

Why 1 : Template separation pad is placing Templates on the conveyor lately.

Why 2 : Sensor used for sensing another Template on conveyor is far, and after the Template passes through the sensor, then only Template pad places next Template on the conveyor.

Why 3 : Sensor Mispositioned.

Root Cause: Mispositioning of the Sensor.

Idea Generation: To change the position of the sensor.

Action: Changed the position of the sensor to near the Template pad.

Countermeasure

SN	Counter Measure Activities	Target Date	Responsibility	Status
01	To study & evaluate all the processes of Machine.	05/12/2018	Maintenance & Production Department	Completed
02	To change the position of sensor from after Oiling system to before oiling system & Make changes in PLC.	08/12/2018	Maintenance Department	Completed
03	To reduce the Machine speed from 16 seconds to 15.5 seconds.	08/12/2018	Production Department	Completed

After Situation

- Changed the position of the sensor to near the Template pad. So that, as soon the Template passes through the sensor, immediately Template pad puts the next Template on the conveyor. By this cycle time of the Template supply system is reduced from 16 seconds to 15 seconds.

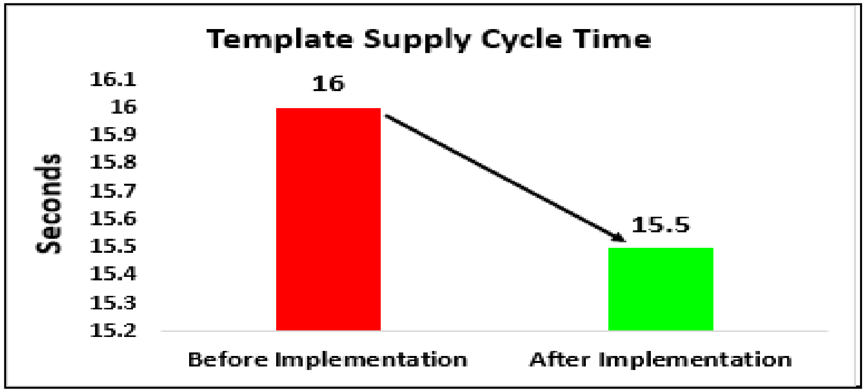
After Situation

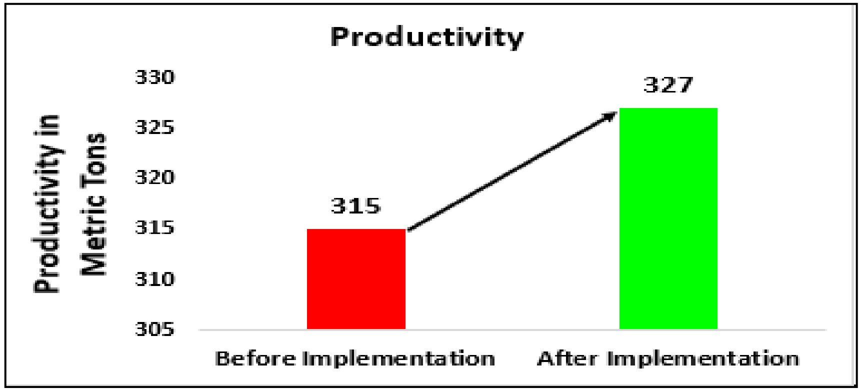
EVEREST

- Changed the position of the sensor to near the Template pad. So that, as soon the Template passes through the sensor, immediately Template pad puts the next Template on the conveyor. By this cycle time of the Template supply system is reduced from 16 seconds to 15 seconds.

Validation of Data: Calculation

Before		After	
Machine Cycle Time	16 Seconds	Machine Cycle Time	15.5 Seconds
No. of Sheets Produced in an Hour	225 sheets	No. of Sheets Produced in an Hour	232 sheets
No. of Sheets Produced Per Day	5400 sheets	No. of Sheets Produced Per Day	5574 sheets
Total Tonnage Per Day	339.3 M.Tons	Total Tonnage Per Day	350.8 M.Tons
Monthly Average Tonnage	315 M.Tons	Monthly Average Tonnage	327 M.Tons





Validation of Permanent Corrective Actions

EVEREST

Tangible Benefits

- Machine Speed has reduced from 16 seconds to 15.5 seconds.
- Monthly Average Productivity has increased by 10 Metric Tons.
- Increase in Revenue of 3.88 Cr./ Annum.
- Potent Cost saving of 1.62 Cr./Annum.

In-tangible Benefits

- Morale of our People increased due to increase in Productivity.
- Management recognized this kaizen as a simple & very effective improvement.

Standardization & Horizontal Deployment

EVEREST

- Training on Updated Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.

De-piling machine Standard Operating Procedure – Updated

1. On the starting of each shift check all the sensors alignment & check for any abnormality in De-piling machine.
2. Piling machine Board operator separates Sheets & Templates by using Sheets separation pad & Template separation pad.
3. De-piled sheets are stacked separately.
4. And De-piled Templates are sent through Roller conveyor to Piling machine.
5. After De-piling Templates, Templates are kept on Roller conveyor by detecting the presence of another template on the conveyor with the help of sensor which has been changed its position before oiling system.
6. So as soon as the template passes through the sensor, Template pad puts next template on the roller conveyor. By which our Template supply cycle time has reduced.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Horizontal Deployment:

BW	CW	KW	LW	PW1	PW2	SW

- Not Applicable - Applicable & Deployed

Kaizen Sheet

Kaizen Sheet — EVEREST

Company	MM/YY	S.No
Everest Industries Ltd	12/18	1

Productivity | Quality | Delivery | Safety | Material | Energy | Maintenance

Kaizen Title: To improve productivity by reducing Machine cycle time.

Implemented Area:
PW1 Shopfloor

Problem/Present Status:
Cycle time of the Template supply system to Piling machine is 16 seconds. Because of this we can't able to reduce the machine speed below 16 sec.

Before Improvement:

Root Cause Identification:

- Problem: •To reduce cycle time of Templates supply system.
- Why 1: •Template separation pad is placing templates on the conveyor lately.
- Why 2: •Sensor used for sensing another template on conveyor is far, and after the template passes through the sensor, then only template pad places next template on conveyor.
- Why 3: •Sensor Mispositioned.

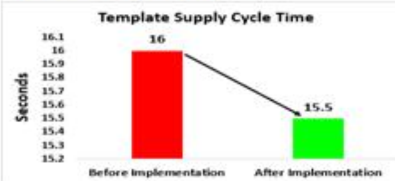
After Improvement:

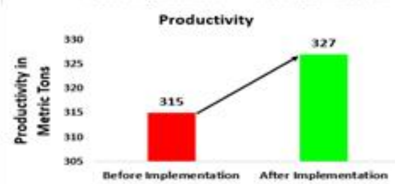
Implemented by:
M. Vijay Kumar, Srinivasagam, M. Gnanaprakash, N. Anil kumar, N. Karthikeyan, Narayanasamy

Result/Benefit:

(a) Qualitative
On-Time supply of Templates to machine without delay.

(b) Quantitative





Root cause	Mispositioning of the Sensor.
Idea to eliminate root cause	To change the position of the sensor.
Counter-measure	Changed the position of the sensor to near the template pad. So that, as soon the template passes through the sensor, immediately Template pad puts the next template on the conveyor. By this cycle time of the Template supply system is reduced from 16 Seconds to 15.5 Seconds.

Standardization:
Preventive Maintenance as per schedule.

Horizontal Deployment

BW	CW	KW	LW	PW	SW

Thank You